

Stranger Forces

Enhancing Virtual Production with Gen-AI in Real-World College Sponsored Studio Collaborations

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Abstract

In spring 2025, NCSU Media Arts, Design and Technology (MADTech) faculty began teaching our students the targeted use of generative AI tools to solve real-world design problems in sponsored studio projects within our College of Design Virtual Production Lab. As presented in our SIGGRAPH SOIREE 2026 fast-forward video “Exploring Gen-AI Tools in a Virtual Production Pipeline”, we hosted Colombian American creative studio Alecardo (Alex Restrepo and Ricardo Tobon) from Los Angeles, and used gen-AI tools to both reduce costs and speed up development of the collaborative sci-fi short film *Las Fuerzas Extrañas* (Strange Forces, 2025). Experimenting with gen-AI tools throughout the virtual production process, including worldbuilding, real-time creature performance and VFX, allowed us to reduce costs and speed up development while achieving Alecardo’s vision. With only a \$15K budget sponsored by a College of Design grant and less than one week to shoot in our lab, students received real-world experience working with professional artists that had worked on blockbuster films and AAA games in the industry. In addition, our hybrid approach to using AI did not eliminate artistic roles in the virtual production pipeline, but it reduced production time and increased visual quality of the final shots. Tobon stated as keynote speaker at our end-of-the-year MADTechFest that he had never completed a professional quality project in such a short time span. We have since applied gen-AI tools for developing sponsored virtual production shots promoting esports for Visit Raleigh in fall 2025 semester, and on collaborative screen dance projects with faculty from the NCSU Performing Arts and Technology department in spring 2026. We discuss how faculty are exploring an ethical hybrid-AI approach to virtual production to better support human artists in achieving their creative visions and prepare students for a changing industry where such technologies are becoming adopted by professional studios and embedded in virtual production pipelines.

CCS Concepts

• Applied computing; • Education; Arts and humanities; • Media arts;

Keywords

Virtual Production, Generative AI, Hyper-Reality, College-Industry Collaboration

ACM Reference Format:

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1 Introduction

Media Arts, Design and Technology (MADTech) faculty taught students the targeted use of generative AI tools for enhancing virtual production shots in the award-winning short film *Las Fuerzas Extrañas* (2025) as part of a real-world NCSU College of Design sponsored collaboration with Colombian American creative studio Alecardo (Alex Restrepo and Ricardo Tobon), as shown in our SIGGRAPH SOIREE 2026 Fast-forward video “Exploring Gen-AI Tools in a Virtual Production Pipeline”. In the spring 2025 semester, Alecardo co-founder Ricardo Tobon was awarded a \$15K virtual production fellowship by the NCSU College of Design Virtual Production Lab, enabling us to host Tobon and Restrepo for a 4-day virtual production shoot in our Raleigh campus lab. Alecardo had adapted the first Latin American sci-fi novel by Leopoldo Lugones, *Las Fuerzas Extrañas* (1908) or *Strange Forces*, to a fantastical short film treatment featuring a giant talking octopus held captive in an abandoned tropical lab. With such a limited production schedule and budget, MADTech faculty and students explored the use of gen-AI tools to speed up and reduce costs throughout virtual production, while ensuring that artistic craft was central to the process and the quality of each shot was maintained for realizing the director’s vision.

For preproduction, graduate students used Blockade Labs AI Skybox to generate a variety of equirectangular images based on Alecardo’s concept art for the abandoned tropical lab. Faculty used Magnific AI’s Mystic tool to creatively upscale and re-style selected background images to make them more hyper-realistic and detailed. We then converted the final background approved by Alecardo to a high dynamic range (HDR) image to import into Unreal Engine 5.4 for worldbuilding with free 3D assets from the FAB Unreal Marketplace. Then in production we used RADiCAL Motion Livelihood streaming to puppeteer the octopus rig by connecting it to a hidden MetaHuman in the UE5 scene. We used a single webcam to spatially perform the creature and trigger virtual lights that indicated the emotional state of the character. The real-time control of the octopus gave the live actress playing the scientist role, performed by Restrepo, a virtual acting partner to react to while we shot in the volume. Both graduate and undergraduate MADTech students helped in developing sets and capturing cinematography



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Table 1: Metadata in tabular format.

Summary	Students learned how to apply hyper-reality design concepts with generative AI tools in sponsored virtual production collaborations with professional clients.
Learning Outcomes	Students learn to integrate the targeted use of gen-AI tools in each phase of the virtual production pipeline to solve real-world design problems according to client specifications.
Classification(s)	Applied computing, Arts and Humanities, Media Arts.
Audience	Undergraduate and graduate students in media arts, games, virtual production and design studies.
Dependencies	Students could attend shoot days that were not always during their scheduled class time.
Prerequisites	No prerequisites were required.
Strengths	Students gained hands-on experience working with professional clients in exploring gen-AI tools to enhance virtual production.
Weaknesses	Subscription costs of testing many gen-AI tools were prohibitive for widespread student use.
Variants	Students could test the tools on their own scenes and projects before working with the sponsored clients.
Assessment	Students were assessed through weekly participation in guest lectures and in-class project development.

during the 4-day shoot as seen in this behind-the-scenes video: <https://youtu.be/9eNF0HGqdlA?si=fwsHrnHlwRYWVZaM>

In postproduction, we used Higgsfield AI to create VFX during the climax when the researcher is obliterated from a storm of cosmic particles unleashed by the angry octopus creature. Using source frames shot in the volume as start and end frames, faculty developed a slow-motion disintegration effect in Higgsfield that could be seamlessly inserted into the footage. This would have been a time consuming and expensive VFX shot to do with standard CGI methods, but Higgsfield generated an approved sequence with surprisingly few iterations. Throughout the development process, students applied Hickman's hyper-reality immersive design concepts they learned in guest lectures and his textbook [Hickman, 2023], and we applied Craig's super-puppet concept to the creature performance that we presented in SIGGRAPH 2025 Spatial Storytelling program [Maraffi, 2025]. Since this project, we have applied gen-AI to create hyper-reality VFX in esports promotional videos for Visit Raleigh in a fall 2025 sponsored special topics course in our lab, where we made a downtown Raleigh mural in our LED volume come alive with animated art. We also began exploring AI for post-production VFX applications for screen dance collaborative projects with faculty from NCSU Performing Arts and Technology Department.

2 Course materials

Click this link to view Alecardo's *Las Fuerzas Extrañas* (2025): <https://youtu.be/cBzl-jSJ460?si=wJrPtK9OEOl5Zbcq> Please contact Topher Maraffi for course syllabi and additional demos of student hybrid-AI work in virtual production.

3 Results and Discussion

Exploring gen-AI tools in our pipeline increased the speed and quality of virtual production shots while retaining creative quality control when using an iterative hybrid approach where human artists guide and evaluate each step. In Tobon's keynote speech at our MADTechFest event at the end of the spring 2025 semester, he said that he had never completed a project of this quality in such

a short time at industry studios like Sony, Dreamworks and Weta. We believe gen-AI tools contributed to the success of this project in achieving Alecardo's vision and receiving placements and awards at international film festivals, including a 2026 Telly Bronze Award for innovative "Craft-Use of Real-Time Rendering". Since developing Alecardo's short sci-fi film, hybrid-AI approaches to virtual production are becoming more adopted by industry studios. Innovative Dreams, a new studio founded by filmmaker Jon Erwin in partnership with Luma AI and Amazon Web Services, demonstrated a similar "realtime hybrid filmmaking" approach that combines performance capture with virtual production using Luma's gen-AI tools [Erwin, 2026]. They believe that this approach is the "future of filmmaking" and that it will create more opportunities for artists and performers in Hollywood [Boorstin et al., 2026]. As gen-AI companies increasingly partner with studios, tools are gaining more embedded affordances for creative inputs beyond prompting, which puts the artist's craft directly into the generative process. Therefore, we continue to ethically explore these emerging technologies in our virtual production projects to prepare our MADTech students for a rapidly changing industry.

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